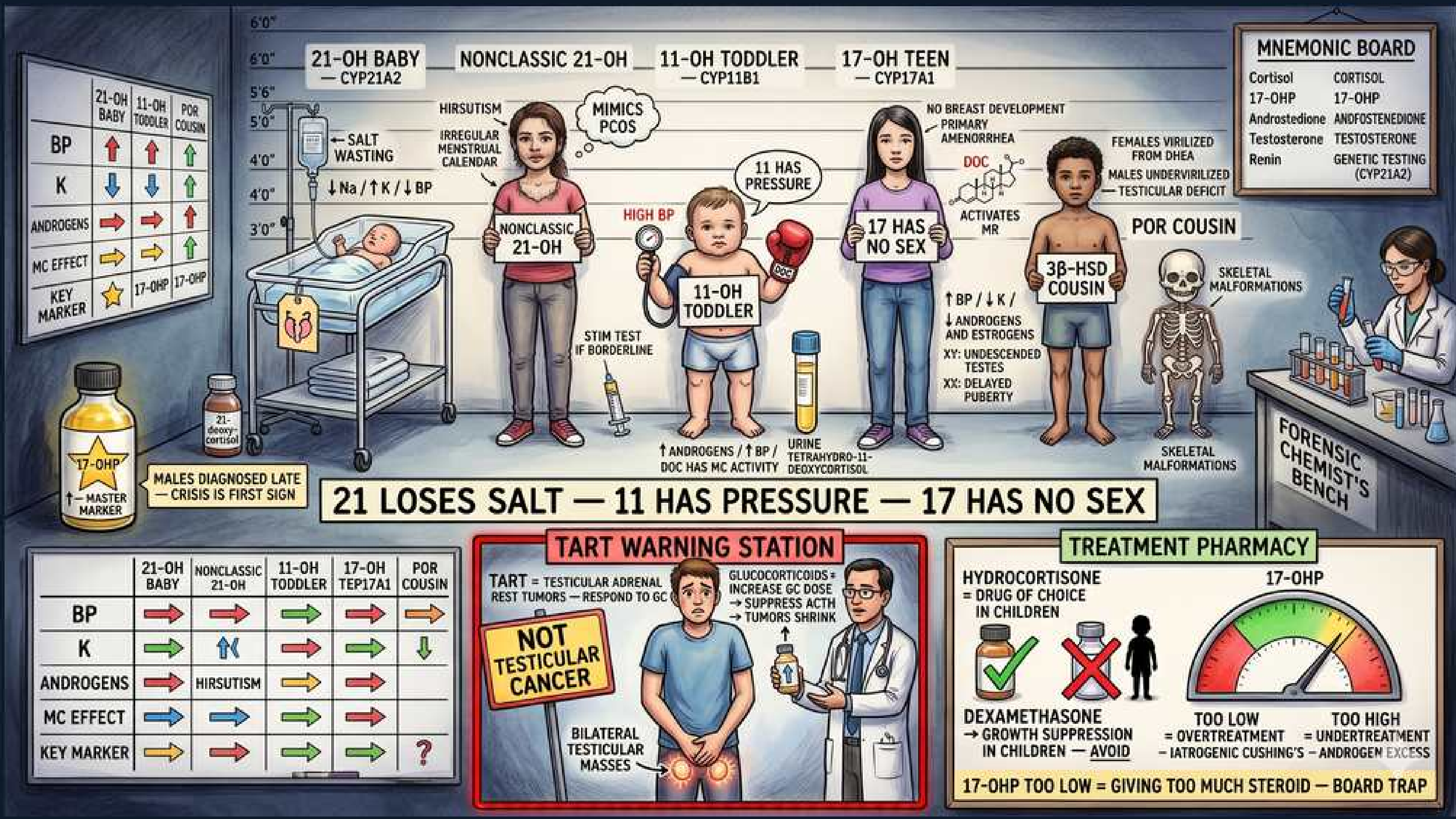
Adrenal Insufficiency/CAH — CAH Triplets (21/11/17)



1. Shared rule banner

WHAT YOU SEE

Central banner: 21 loses salt, 11 has pressure, 17 has no sex

WHAT IT ENCODES

All CAH are autosomal-recessive enzyme blocks in steroidogenesis. The shared signature is LOW cortisol → loss of negative feedback → HIGH ACTH → bilateral adrenal HYPERPLASIA and pile-up of precursors above the block. They differ only by where the precursors get shunted, which sets BLOOD PRESSURE and ANDROGENS. Mnemonic: 21 = salt loss (hypotension), 11 = high pressure (HTN), 17 = no sex (no puberty).

BOARD TRAP

All three share low cortisol + high ACTH; the BP and androgen status are what let you name the specific enzyme.

2. 21-OH master marker

WHAT YOU SEE

Bottle labeled 17-OHP master marker on the left

WHAT IT ENCODES

17-hydroxyprogesterone (17-OHP) is the substrate just proximal to 21-hydroxylase, so it accumulates massively when that enzyme is blocked — it is the diagnostic MASTER MARKER and the analyte measured on newborn screening. 21-hydroxylase deficiency (CYP21A2) is by far the most common CAH (~90–95% of cases).

BOARD TRAP

A markedly elevated 17-OHP is essentially diagnostic of 21-hydroxylase deficiency specifically — not the other enzyme blocks.

3. 21-OH salt-wasting baby

WHAT YOU SEE

Newborn in bassinet labeled 21-OH, salt wasting

WHAT IT ENCODES

Classic salt-wasting 21-OH CAH blocks BOTH cortisol and aldosterone synthesis: low aldosterone → renal salt loss → HYPOnatremia, HYPERkalemia, volume depletion/hypotension, metabolic acidosis. It presents in the first 1–2 weeks of life as a salt-wasting crisis (vomiting, poor feeding, shock) — a neonatal emergency.

BOARD TRAP

Salt-wasting 21-OH gives LOW Na / HIGH K (no mineralocorticoid) — the exact OPPOSITE of the high-BP/low-K picture of 11- and 17-hydroxylase deficiency.

4. Female virilization

WHAT YOU SEE

21-OH baby chart showing high androgens / ambiguous genitalia

WHAT IT ENCODES

With cortisol synthesis blocked, precursors are shunted into the androgen pathway → androgen EXCESS. Genetic (46,XX) females are born virilized with ambiguous/clitoromegaly genitalia; affected boys appear normal at birth but may show later precocious puberty. CAH is the most common cause of ambiguous genitalia in the newborn female.

BOARD TRAP

A virilized newborn girl who then crashes with hyponatremia/hyperkalemia = classic salt-wasting 21-OH CAH.

5. Nonclassic 21-OH

WHAT YOU SEE

Adolescent labeled nonclassic 21-OH, mimics PCOS

WHAT IT ENCODES

Nonclassic (late-onset) 21-OH CAH has partial enzyme activity: cortisol and aldosterone are near-normal (no salt-wasting), and patients present later with hyperandrogenism — hirsutism, acne, irregular menses, infertility — closely MIMICKING PCOS. Baseline 17-OHP may be only mildly raised, so a cosyntropin (ACTH) stimulation test is used to confirm.

BOARD TRAP

Nonclassic CAH is NOT salt-wasting (cortisol/aldo near-normal); think of it as the PCOS mimic and confirm with an ACTH stim test when baseline 17-OHP is borderline.

6. 11-OH high BP toddler

WHAT YOU SEE

Toddler with boxing glove labeled 11-OH, 11 has pressure

WHAT IT ENCODES

11-beta-hydroxylase deficiency (CYP11B1) blocks the final step to cortisol, so 11-deoxycorticosterone (DOC) and 11-deoxycortisol pile up. DOC is a potent MINERALOCORTICOID → salt/water retention → HYPERTENSION with HYPOkalemia (and low renin). Precursors still feed androgens → virilization. It is the 2nd most common CAH.

BOARD TRAP

11-OH deficiency causes HIGH BP + low K (DOC excess) — it is a salt-RETAINING, not salt-wasting, form despite also virilizing.

7. DOC buildup

WHAT YOU SEE

DOC / 11-deoxycortisol precursors stacked near the 11-OH toddler

WHAT IT ENCODES

In 11-OH deficiency, the accumulating 11-DEOXYCORTICOSTERONE (DOC) acts on the mineralocorticoid receptor → Na^+/water retention, K^+ loss and HTN, while renin and aldosterone themselves are suppressed. Elevated 11-deoxycortisol and DOC are the biochemical fingerprints.

BOARD TRAP

The hypertension in 11- (and 17-) OH deficiency is driven by DOC, not aldosterone — so aldosterone and renin are actually LOW.

8. 17-OH teen no sex

WHAT YOU SEE

Adolescent labeled 17-OH, 17 has no sex, low androgens/estrogens

WHAT IT ENCODES

17-alpha-hydroxylase deficiency (CYP17A1) blocks both cortisol AND all sex steroids while shunting into the mineralocorticoid (DOC) pathway: HYPERTENSION + HYPOkalemia, LOW cortisol, and LOW androgens/estrogens → absent/delayed puberty. 46,XX girls have primary amenorrhea; 46,XY boys are under-virilized (may present phenotypically female).

BOARD TRAP

17-OH deficiency is the only CAH that UNDER-virilizes (no sexual development) AND causes HTN — high BP + no puberty is the buzz combination.

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9. 3-beta-HSD cousin

WHAT YOU SEE

Figure labeled 3-beta-HSD cousin near skeleton

WHAT IT ENCODES

3-beta-HSD type 2 deficiency blocks conversion of pregnenolone-class steroids, so only the WEAK androgen DHEA accumulates: 46,XX females are mildly virilized (DHEA), while 46,XY males are UNDER-virilized (can't make testosterone). Often a salt-wasting picture too since aldosterone synthesis is impaired.

BOARD TRAP

3-beta-HSD is the one where the boy is under-virilized but the girl is mildly virilized — opposite-direction effects from a weak androgen.

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10. POR cousin

WHAT YOU SEE

Skeleton labeled POR cousin with skeletal malformations

WHAT IT ENCODES

P450 oxidoreductase (POR) deficiency impairs the electron donor shared by 21-hydroxylase, 17-hydroxylase and aromatase simultaneously → a mixed/combined biochemical picture plus the skeletal Antley-Bixler malformations (craniosynostosis, radiohumeral synostosis).

BOARD TRAP

CAH presenting with skeletal/craniofacial malformations (Antley-Bixler) points to POR deficiency.

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11. TART warning station

WHAT YOU SEE

Red box TART warning station, NOT testicular cancer, bilateral testicular masses

WHAT IT ENCODES

Chronically poorly controlled CAH keeps ACTH high; ACTH stimulates ectopic adrenal-rest tissue in the testes → testicular adrenal rest tumors (TARTs). They are BENIGN, typically BILATERAL, can impair fertility, and SHRINK with better glucocorticoid suppression of ACTH — so the fix is more steroid, not surgery.

BOARD TRAP

Bilateral testicular masses in a CAH patient are TART (ACTH-driven adrenal rests), NOT testicular cancer — treat by intensifying steroid suppression, do not orchiectomize.

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12. Hydrocortisone tx

WHAT YOU SEE

Treatment pharmacy: hydrocortisone, drug of choice in children

WHAT IT ENCODES

Hydrocortisone is the glucocorticoid of choice in growing children with CAH: it both REPLACES cortisol and suppresses the ACTH drive that fuels androgen excess. Its short half-life minimizes growth suppression. Salt-wasting (21-OH) patients also need fludrocortisone ± sodium chloride supplementation in infancy.

BOARD TRAP

Avoid long-acting dexamethasone/prednisone in growing children — they cause growth suppression and iatrogenic Cushing; use hydrocortisone.

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13. 17-OHP monitoring gauge

WHAT YOU SEE

Gauge: 17-OHP too low = overtreatment (Cushing), too high = undertreatment

WHAT IT ENCODES

Titrate therapy by 17-OHP and androstenedione/testosterone, NOT by normalizing 17-OHP fully. A suppressed/too-low 17-OHP signals OVER-replacement → iatrogenic Cushing, growth suppression and weight gain; a too-high 17-OHP signals UNDER-treatment → persistent androgen excess, virilization and accelerated bone age.

BOARD TRAP

Driving 17-OHP all the way to zero/normal is the classic trap — it means you've over-suppressed and risk iatrogenic Cushing; aim for slightly-elevated, not normal.
